

The Number System

Week 3

Topics

1. Binary Numbers
2. Octal Numbers
3. Hexadecimal Numbers
4. Converting Decimal to Binary and Vice versa
5. Converting Decimal to Octal and Vice versa
6. Converting Decimal to Hexadecimal and Vice versa
7. Converting Binary, Octal and Hexadecimal Numbers

Objectives

- Know and demonstrate how to convert number from one system to another:
 - Decimal to Binary to Decimal
 - Binary to octal to binary
 - Binary to hexadecimal to binary
 - Octal to Decimal to Octal
 - Octal to Hexadecimal to Octal
 - Hexadecimal to Decimal to Hexadecimal

Decimal Number System

- Decimal number system – is a positional numeral system employing 10 as the base & has 10 different numerals: 0, 1, 2, 3, 4, 5, 6, 7, 8 & 9

Example:

$$543.21 = (5 * 10^2) + (4 * 10^1) + (3 * 10^0) + (2 * 10^{-1}) + (1 * 10^{-2})$$

Decimal Number System

$$3000_{10} = 3 * 10^3$$

Radix/Base

Exponent

1. Radix and "weight"

- Decimal numbers' "weight" and its meaning
 - "10" is called "Radix"
 - upper right of 10 (in this example, 3) is called "exponent"
- 3 is the coefficient
- 10 is the radix
- 3 is the exponent

Decimal Number System

$$\begin{aligned}(1456)_{10} &= 1000 + 400 + 50 + 6 \\&= 1*1000 + 4*100 + 5*10 + 6 \\&= 1*10^3 + 4*10^2 + 5*10^1 + 6*10^0\end{aligned}$$

$$\begin{aligned}(0.25)_{10} &= 2/10 + 5/100 \\&= 2/10^1 + 5/10^2 \\&= 2*10^{-1} + 5*10^{-2} \\&= 0.2 + 0.05 \\&= 0.25\end{aligned}$$

Binary, Octal and Hexadecimal

Binary numbers – A base 2 number system that uses only two digits: 0 and 1.

Octal numbers – A base 8 number system that uses only seven digits: 0, 1, 2, 3, 4, 5, 6, 7

Hexidecimal - A base 16 number system that uses only 16 digits & symbols: – 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A (10), B (11), C (12), D (13), E (14), and F (15)

Examples:

Binary: B(101) = (101)₂

Ocat: O(701) = (701)₈

Hex: H(A23) = (A23)₁₆

Binary Number System

- A base 2 number system that uses only two digits: 0 and 1
- The equivalent binary value of all decimal numbers should be expressed in terms of 0 and 1, as shown in the table.

Decimal	Binary
0	0
1	1
2	10
3	11
4	100
5	101
6	110
7	111
8	1000
9	1001

Why Study Binary Number System?

- The different parts of a computer communicate through pulses of current (1s and 0s).
- It is the primary language of computing systems
- All types of information, such as numbers, text, and images, are using binary code
- understanding binary is essential for working with computer programming languages and developing software applications.
- Binary system form the basis for designing and building digital circuits and systems that perform various computational tasks

Decimal	Binary
0	0
1	1
2	10
3	11
4	100
5	101
6	110
7	111
8	1000
9	1001

Octal Number System (OCT)

Decimal	BIN	OCT
0	000	0
1	001	1
2	010	2
3	011	3
4	100	4
5	101	5
6	110	6
7	111	7
8	1000 or 001 000	10
9	1001 or 001 001	11

- A base-8 number system that uses the digits 0, 1, ... 7.
- Provides a convenient way of converting large binary numbers into more compact and smaller groups.
- Each octal number represents 3 binary numbers.

Octal Number System (OCT)

Decimal	BIN	OCT
0	000	0
1	001	1
2	010	2
3	011	3
4	100	4
5	101	5
6	110	6
7	111	7
8	1000 or 001 000	10
9	1001 or 001 001	11

- In the earlier days of computing, octal numbers was very popular for counting inputs and outputs (I/Os) because they work in counts of eight, a byte at a time.
- These days, it is used less frequently than the HEX numbering system.

Hexadecimal Number System (HEX)

Decimal	BIN	HEX
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

- A 16-based number system that uses numbers 1-10 and A-F/ a-f to 10-15.
- Each HEX digit represents 4 bits
- The primary use of HEX notation is a human-friendly representation of binary coded values in computing and digital electronics.

Hexadecimal Number System (HEX)

Decimal	BIN	HEX
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

- Byte values in the range of 00000000 to 11111111 can be represented more conveniently as two hexadecimal digits in the range 00 to FF.
- Hexadecimal is also commonly used to represent computer memory addresses.

Decimal	BIN	OCT	HEX
0	0	0	0
1	1	1	1
2	10	2	2
3	11	3	3
4	100	4	4
5	101	5	5
6	110	6	6
7	111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

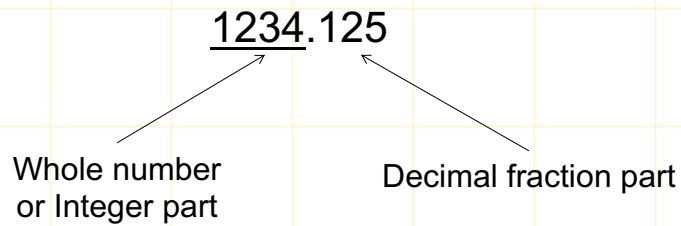
Radix/Base Conversion

- To process numeric values in a computer, decimal numbers are converted into binary, octal (oct) or hexadecimal (hex) numbers
- This operation is called radix/base conversion as follows:
 1. Decimal to Binary, Octal and Hexadecimal
 2. Binary to Decimal, Octal, and Hexadecimal
 3. Octal to Decimal, Binary and Hexadecimal
 4. Hexadecimal to Binary, Decimal and Octal

Decimal Numbers – Binary Conversion

Converting Decimal Numbers to Binary

Convert Decimal Number 1234.125 to Binary number



Converting the integer part is done separately with the decimal part.

Converting Decimal Numbers to Binary

Steps in converting the integer part

1. Divide the integer part into 2 using the “upside down” long division symbol. Write the quotient under the long division symbol and the remainder at the right side.

$$\begin{array}{r} 2 \overline{) 1234} \quad \text{Remainder} \\ 2 \overline{) 617} \quad 0 \end{array}$$

Converting Decimal Numbers to Binary

2) <u>1234</u>	Remainder
2) <u>617</u>	0
2) <u>308</u>	1
2) <u>154</u>	0
2) <u>77</u>	0
2) <u>38</u>	1
2) <u>19</u>	0
2) <u>9</u>	1
2) <u>4</u>	1
2) <u>2</u>	0
2) <u>1</u>	0
0	1

3. Divide the quotient again by 2 and repeat the same process until the quotient becomes 0

4. Write out the equivalent binary number starting with the bottom remainder upwards to the top.

5. Therefore, the equivalent of 1234 is $(10011010010)_2$

Converting Decimal Numbers to Binary

Converting the decimal part of 1234.125

1. Multiply the decimal fraction by 2,
2. Get the integer part of the [result](#) and write it as the number after the period of the equivalent binary number.
3. Apply the above process to the remaining decimal part of the original number until it becomes 0.

$$0.125 * 2 = \underline{0}.25 \rightarrow (0.125)_{10} = (0.\underline{0})_2$$

$$0.25 * 2 = \underline{0}.5 \rightarrow (0.125)_{10} = (0.\underline{00})_2$$

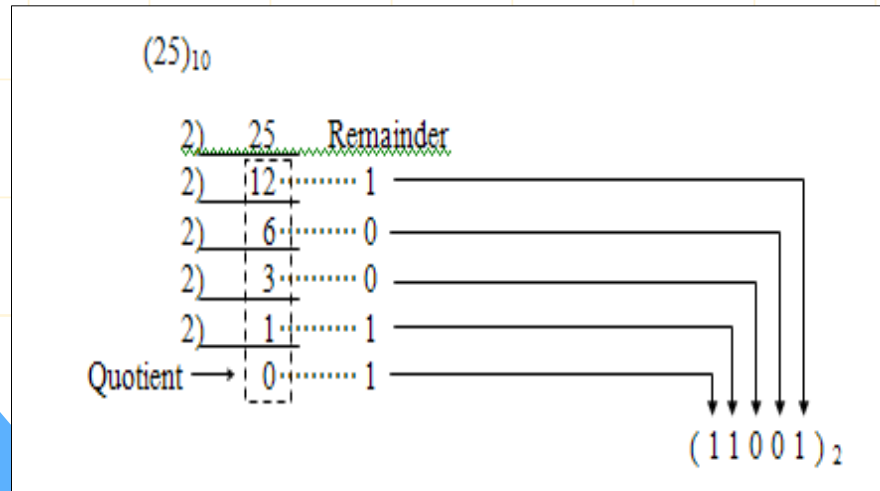
$$0.5 * 2 = \underline{1}.0 \rightarrow (0.125)_{10} = (0.\underline{001})_2$$

$$0.0 * 2 = 0$$

So, $(1234.125)_{10} = (10011010010.001)_2$

Converting Decimal Numbers to Binary

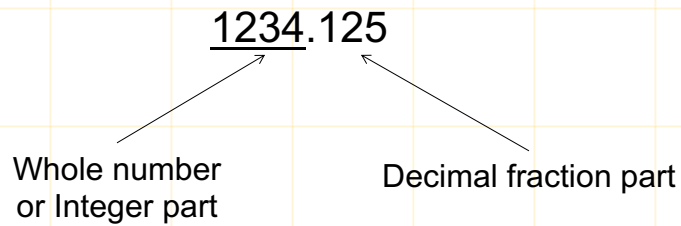
Example 2:



Decimal – Octal Conversion

Decimal – Octal Conversion

Convert Decimal Number 1234.125 to Octal number



Again, the process of converting the integer part is done separately with the decimal part.

Decimal – Octal Conversion

Steps:

1. Divide the integer part into 8 using the “upside down” long division symbol.
2. Write the quotient under the long division symbol and the remainder at the right side.

8	<u>1234</u>	Rem	
8	<u>154</u>	2	
8	<u>19</u>	2	
8	<u>2</u>	3	
	0	2	

↑

Decimal – Octal Conversion

8 | 1234 Rem

8 | 154 2

8 | 19 2

8 | 2 3

0 2

3. Divide the quotient again by 8 and repeat the same process until the quotient becomes 0

4. Write out the equivalent OCTAL number starting with the bottom remainder upwards to the top.

5. Therefore, the equivalent of 1234 is $(2322)_8$

Decimal – Octal Conversion

Converting the decimal part of 1234.125

1. Multiply the decimal fraction by 8,
2. Get the integer part of the [result](#) and write it as the number after the period of the equivalent OCT number.
3. Apply the above process to the remaining decimal part of the original number until it becomes 0.

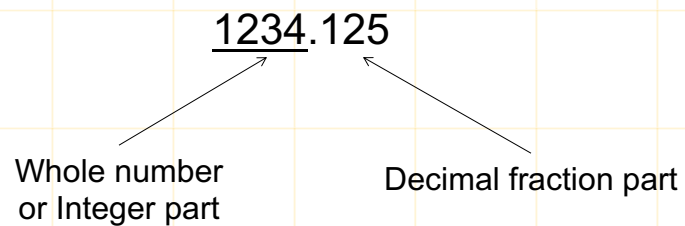
$$0.125 * 8 = \underline{1.00} \rightarrow (0.125)_{10} = (0.\underline{1})_8$$

4. Therefore, $(1234.125)_{10} = (2322.1)_8$

Decimal – Hexadecimal Number Conversion

Decimal – Hexadecimal Number Conversion

Convert Decimal Number 1234.125 to HEX number



Again, the process of converting the integer part is done separately with the decimal part.

Decimal – Hexadecimal Number Conversion

Steps:

1. Divide the integer part into 16 using the “upside down” long division symbol.

16 $\overline{)1234}$ Rem

16 $\overline{)77}$ 2

16 $\overline{)4}$ 13

0 4

2. Write the quotient under the long division symbol and the remainder at the right side.

Decimal – Hexadecimal Number Conversion

16 $\overline{)1234}$ Rem

16 $\overline{)77}$ 2

16 $\overline{)4}$ 13

0 4

3. Divide the quotient again by 16 and repeat the same process until the quotient becomes 0

4. Write out the equivalent Hexadecimal number starting with the bottom remainder upwards to the top.

5. Therefore, the equivalent of 1234 is $(4D2)_{16}$, where $D = 13$.

Decimal – Hexadecimal Number Conversion

Converting the decimal part of 1234.125

1. Multiply the decimal fraction by 16,
2. Get the integer part of the [result](#) and write it as the number after the period of the equivalent HEX number.
3. Apply the above process to the remaining decimal part of the original number until it becomes 0.

$$0.125 * 16 = \underline{2}.00 \rightarrow (0.125)_{10} = (0.\underline{2})_{16}$$

4. Therefore, $(1234.125)_{10} = (4D2.2)_{16}$

Summary:

1. Decimal to Binary

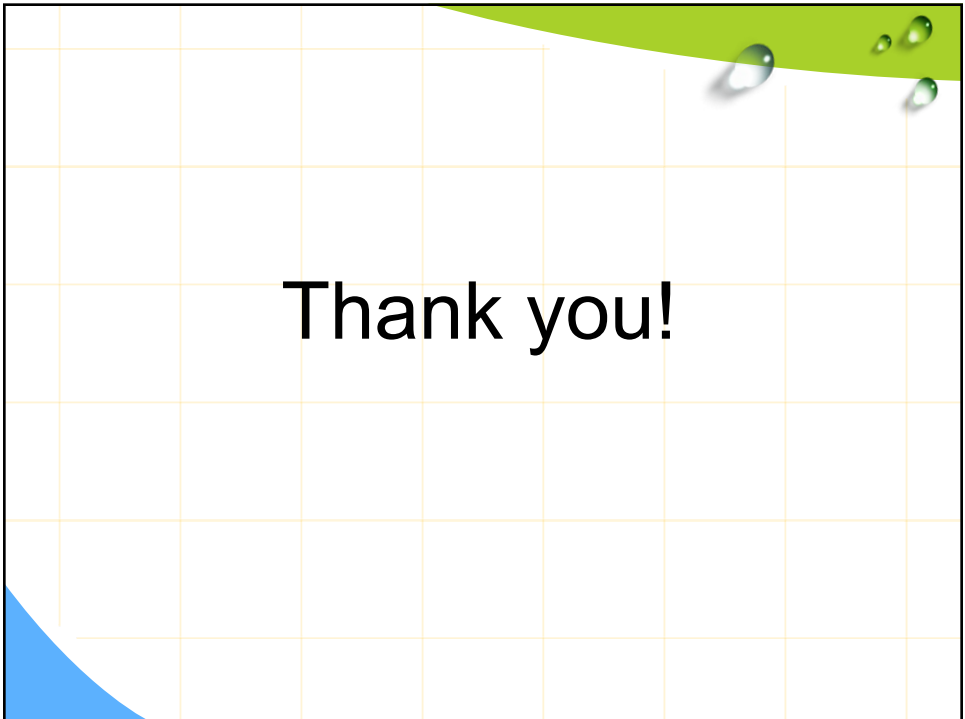
$$(1234.125)_{10} = (10011010010.001)_2$$

2. Decimal to Octal

$$(1234.125)_{10} = (2322.1)_8$$

3. Decimal to Hexadecimal

$$(1234.125)_{10} = (4D2.2)_{16}$$



Thank you!